Soil DNA extraction, semi-automated (SuperSoil, Zixpress 32)

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(v1.1, 2026-10-20; current lab method; appendix for difficult samples)

Source of record: Huang MEGA Lab (NCKU). Bench steps from the lab's protocols.io protocol *SuperSoil DNA Extraction Modified (w. Zixpress 32)* (dx.doi.org/10.17504/protocols.io.yxmvmdy99v3p/v1, created 2026-08-27). Appendix (difficult samples) from the lab note *Soil DNA yield optimization* (2026). This PDF is the lab's self-hosted copy.

Principle: bead-beating lysis in SD1 → inhibitor (humic) precipitation with SD2 → magnetic-bead binding in SD3 on the Zixpress 32 (wash SEA, 2× 80% ethanol, dry, elute 100 µL). Manual steps end at the plate; the instrument does binding → elution. Routine samples: Procedure below. Difficult or low-yield soils: Appendix.

Safety: SEA and ethanol are flammable. Do not add bleach or acid to the sample-preparation waste (thiocyanate).

Procedure

Sample preparation and cell lysis

1. (2 min) Into a 1.5 mL screw tube add 0.2 mL ⌀1.0 mm beads, 0.2 mL ⌀0.5 mm beads, **800 µL** SD1 and **0.25 g** soil. Vortex briefly.

※ **Note:** Loose soil that fills the tube: split into two tubes, run both, and merge the lysates at Step 5.

2. (10 min) Vortex-Genie 2, horizontal holder, maximum speed, **10 min**.

3. (5 min) Centrifuge **14 000 rpm**, **25 °C**, **5 min**.

Inhibitor precipitation

4. Pipette **200 µL** SD2 into a clean 1.5 mL microcentrifuge tube.

5. Transfer **500 µL** supernatant into the SD2 tube; vortex **5 s**. A white precipitate forms.

※ **Note:** Excess SD2 ($AlCl_3$) co-precipitates the DNA with the inhibitors: do not increase SD2. If particles stay suspended, set a P200 to 150 µL and pipette the supernatant up and down 3 times.

6. (2 min) Centrifuge **14 000 rpm**, **25 °C**, **2 min**.

Plate loading (96 deep-well plate, one row per sample)

7. Well 6 (binding): **450 µL** SD3 + **50 µL** magnetic beads (10 mg/mL) + **500 µL** cleared lysate from Step 6 = **1000 µL**.

8. Well 5 (wash 1): **500 µL** SEA.

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9. Wells 4 and 3 (wash 2): **500 µL** 80% ethanol each.

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10. Well 2 (elution): **100 µL** DEPC-treated water. Well 1 stays empty (waste).

Well	Content	Volume
1	Waste	—
2	DEPC-treated water (elution)	**100 µL**
3	80% ethanol	**500 µL**
4	80% ethanol	**500 µL**
5	SEA (wash buffer 1)	**500 µL**
6	Binding mix: SD3 + beads + lysate	**1000 µL (Appendix: 1200 µL)**

Automated extraction

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11. (50 min) Load the plate in the Zixpress 32 and run the **power soil** program. Check the settings match the table; only Well 6 volume ever changes (1000 µL routine, 1200 µL Appendix).

Well	Step	Standby (min)	Mix (min)	Volume (µL)	Mix speed	Mag (s)	Temp (°C)
6	Binding	0	15	1000	3	300	0
5	Wash1	0	5	500	2	120	0
4	Wash2-1	0	3	500	2	120	0
4	Wash2-2	0	3	500	2	120	0
3	Dry	6	0	0	0	0	0
2	Elution	0	10	100	1	120	60
1	Waste	0	1	0	2	0	0

gDNA collection and quantification

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12. Place the plate on the 96 magnetic plate to pellet residual beads.

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13. Collect **100 µL** eluate from Well 2 without carrying beads; this is the DNA template.

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14. Qubit dsDNA HS: **1 µL** DNA + **199 µL** working solution. Register the extraction as a Stage 3a draft in `/lims-register`; the MolWork# is minted after the reading is entered.

※ **Note:** Readings below 0.05 ng/µL sit at the detection floor and are unreliable: repeat with 5 µL input, or cross-check by Nanodrop / gel.

Appendix: difficult or low-yield samples (scaled lysis + second extraction)

Use when: routine extraction gives low yield or the sample is hard to lyse (clay-rich, dry, organic-poor soils). Gave the best yield in the lab's yield-optimization test; not used by default. Record the actual

supernatant volumes at Steps 19 and 22.

Scaled lysis and re-extraction

15. Into a **2.0 mL** screw tube add 0.2 mL ⌀1.0 mm beads, 0.2 mL ⌀0.5 mm beads, **1000 µL** SD1 and **0.25 g** soil.

16. Incubate **65 °C**, **10 min**.

17. Vortex-Genie 2, maximum speed, **10 min**.

18. Centrifuge **14 000 rpm**, **25 °C**, **5 min**.

19. Transfer the supernatant to a clean 2.0 mL tube; record the volume (expect **650–750 µL**).

20. Add **500 µL** SD1 to the original screw tube; vortex at maximum speed **3 min** (rinses the residual liquid phase; extend to 5–10 min only if the first lysis looks incomplete).

21. Centrifuge **14 000 rpm**, **25 °C**, **5 min**.

22. Pool this supernatant into the Step 19 tube; record the combined volume (expect **1000–1200 µL**).

Inhibitor precipitation and plate loading (difficult samples)

23. In a new 2.0 mL tube add SD2 = combined volume × **0.4** (e.g. 1000 µL → **400 µL** SD2). Keep the ratio at 0.4: more SD2 risks precipitating the DNA.

24. Transfer all pooled supernatant onto the SD2; vortex **5 s**. If the total exceeds the 2.0 mL tube (1400–1700 µL possible), split into two tubes, centrifuge both and pool the cleared supernatants.

25. Centrifuge **14 000 rpm**, **25 °C**, **2 min**.

26. Well 6: **450 µL** SD3 + **50 µL** beads + **700 µL** cleared supernatant = **1200 µL**. Wells 1–5 as in the routine procedure. In the power soil program set Well 6 **Volume = 1200**; all other parameters unchanged. Continue from Step 12.

※ **Note:** If this procedure gives A260/230 < 1.5 (humic carry-over), raise the SD2 ratio to 0.5 on the next attempt. If the first supernatant already yields ~750 µL, the second extraction adds little: use 1000 µL SD1 only and skip Steps 20–22.

Materials

#	**Material**
1	Zirconia beads ⌀0.5 mm (CAAR-ZB-0.5-1000BX) and ⌀1.0 mm (CAAR-ZB-1-1000BX), 0.2 mL each per tube
2	**1.5 mL screw-cap extraction tubes (2.0 mL for the Appendix); 1.5 mL microcentrifuge tubes**
3	Solution SD1 (chaotropic lysis): 2.5% sodium thiocyanate, 5% Na_2HPO_4, pH ~8.4 (w/w)
4	Solution SD2 (inhibitor precipitation): 65% guanidinium thiocyanate, pH ~7.0 (w/w); contains $AlCl_3$
5	Solution SD3 (binding): 65% guanidinium thiocyanate, pH ~7.0 (w/w)
6	Solution SEA (wash buffer 1): 50% ethanol, 10% isopropanol, 30% guanidine hydrochloride (w/w)
7	**80% ethanol; DEPC-treated water; magnetic beads 10 mg/mL (hydroxyl silica)**
8	**96 deep-well plate (2 mL square wells); 96 magnetic plate; Zixpress 32**
9	Vortex-Genie 2 with SI-H524 horizontal holder; microcentrifuge; Qubit Flex, dsDNA HS

※ **Note:** If SD1 has precipitated, leave it at room temperature for 30 min; it clears again. Keep the 50 mg/mL "H" bead stock diluted to 10 mg/mL working solution so the protocol volumes stay unchanged.